Measures of Dispersion (AS Level Psychology 9990)

1️⃣ Range

$$Range = Highestvalue - Lowestvalue$$

✅ Explanation:

The **range** is the difference between the highest and lowest score.

📌 Example:

Scores: 4, 6, 7, 10, 15

Range = 15 − 4 = **11**

✓ Advantages:

- Very easy to calculate
- Quick measure of spread

❌ Disadvantages:

- Affected by extreme values
- Only uses two scores (highest & lowest)
- Not very accurate for research conclusions

🧠 In Psychology:

If one participant scores extremely high or low in a memory experiment, the range will increase a lot — even if most scores are similar.

2️⃣ Standard Deviation (SD)

Standard deviation tells us **how far scores deviate (spread) from the mean**.

If SD is:

- **Small** → scores are close to the mean
- **Large** → scores are widely spread

📌 Formula (Concept Form)

$$s = \sqrt{[\Sigma(x - \bar{x})^2/n]}$$

Where:

- $\bar{x}$ = mean
- $x - \bar{x}$ = deviation from mean
- Σ = sum
- n = number of scores

(You usually don't need to calculate this manually in AS exams, but understand what it means.)

📊 Simple Example

Data Set A:

10, 11, 10, 9, 10

Mean = 10
Scores are very close to mean → **Low SD**

Data Set B:

2, 6, 10, 14, 18

Mean = 10
Scores are spread out → **High SD**

Both sets have the same mean, but very different standard deviations.

✓ Advantages:

- Uses all values
- Most accurate measure of dispersion
- Important in inferential statistics

❌ Disadvantages:

- More complex to calculate
- Can be influenced by extreme scores

📈 Why Standard Deviation is Important in Psychology

Researchers prefer **standard deviation** because:

- It shows reliability of results
- Smaller SD = more consistent performance

- Large SD = individual differences are high

Example:
If a therapy reduces anxiety and has **low SD**, it means most participants improved similarly.

📘 Comparison: Range vs Standard Deviation

Feature	Range	Standard Deviation
Uses all scores?	❌ No	✅ Yes
Affected by outliers?	✅ Yes	✅ Yes (but less extreme)
Accuracy	Low	High
Ease of calculation	Very easy	More complex

📝 Exam-Style 6 Mark Answer

Question:
Describe measures of dispersion. (6 marks)

✅ Sample A-Grade Answer:

Measures of dispersion show how spread out data is around the mean. The two main measures are range and standard deviation.

The range is calculated by subtracting the lowest score from the highest score. It is simple to calculate but is affected by extreme values and only uses two scores.

Standard deviation measures how much scores deviate from the mean. It uses all values in the data set and provides a more accurate measure of variability. A small standard deviation

indicates scores are clustered closely around the mean, while a large standard deviation indicates scores are widely spread.

Standard deviation is preferred in psychological research because it gives a more reliable measure of variability.